Università degli Studi di Palermo

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Paper 1: response to the internal review

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To: Prof. Mauro De Marchis 13 October 2026

Each of the five observations is recorded below with what was found on checking it against the manuscript, so that the scope of each change is known before it is made. None has been implemented yet.

Two further sections follow. The first carries the sections not yet reviewed, scanned against the criterion the five observations share. The second measures the manuscript against the journal's stated requirements.

1. The five observations

1. Carmelo Nasello and Antonino Maltese as co-authors

Three details are not settled by the observation and would benefit from a decision: position in the author order, affiliation where it differs from Dipartimento di Ingegneria, and the CRediT roles. The three current entries follow the pattern of the Sentinel-1 reservoir manuscripts, where supervising co-authors carry Conceptualization, Supervision, Methodology and Writing (review and editing). Antonino Maltese supplied the classification method that Section 3.4 builds on, which may warrant a role beyond that pattern.

Elsevier does not permit authorship changes after acceptance, and changes before acceptance require written agreement from every author, so the list should be definitive at submission.

2. Lagoon geometry in the Introduction

The duplication is complete. The second Introduction paragraph and Section 2 carry the same quantities: 11 km north to south, 2.5 km east to west, 2200 ha, the northern inlet at 400 m wide and 0.30–0.40 m deep, the southern at 2900 m and 1.0–1.5 m, and shoot densities of 1000–1200 plants m^{-2}. The figures can therefore be cut from the Introduction without loss.

Two elements of that paragraph are argument rather than description, and a decision on where they belong is separate from the geometry: the sub-metre mean depth, which is the ground for rejecting the depth-averaging assumption, and the 75% reduction in *P. oceanica* extent over 2003–2024, which motivates the vegetated arm of the design.

3. Age of the reference set

Twenty-two of the 29 entries predate 2015, and the Introduction draws on nineteen works of which most fall in that group.

The set divides between the formulations the model implements and the claims that frame the problem. Booij et al. (1999), Baptist et al. (2007), Kernkamp et al. (2011) and Liu and Weisberg (2011) are cited because the model implements those specific formulations, and a newer reference does not substitute for them. The candidates for refreshment are the framing claims, where a recent synthesis can stand beside or in place of an older source, and the Mediterranean lagoon context presently carried by Niedda and Greppi (2007) and Umgiesser et al. (2014).

A forward-citation search on each older work, filtered to coastal lagoon hydrodynamics, seagrass canopy flow and Lagrangian validation, is the method proposed.

4. Figure 1

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The margin is in panel (a), between the map and the axes frame, and the panel takes its limits from a constant that does not match the mesh it paints:

	Panel (a) limits	Mesh extent	Margin
Longitude	11.950 to 12.570	11.9600 to 12.5581	0.0100 W, 0.0119 E
Latitude	37.690 to 38.072	37.7000 to 38.0631	0.0100 S, 0.0089 N

Panel (b) takes its limits from the lagoon bounding box and paints to the frame, so no margin appears there. The latitude band in panel (a) is 2.6% of the panel range, which accounts for the millimetre-scale difference in apparent map height. Matching panel (a) to the mesh extent removes it.

5. Justification of the choices in Section 3.1

Section 3.1 states the mesh, the ten sigma layers, the eight partitions and the time step without the ground for each. Two can be supplied immediately. The partition count is a throughput choice rather than a physical one, and the supporting measurement exists, the eight-rank configuration reaching about nine hours per nine simulated days on the 64-core machine; saying so removes the implication that eight carries physical meaning. The discretisation method is documented in the project record, from the Navier–Stokes equations to the discrete system the solver advances, and Section 3.1 can draw on it.

The mesh resolution and the layer count are different. Neither has a convergence test behind it. A defensible treatment is to state the constraint each choice answers and to record that no sensitivity test was run, rather than to imply a convergence study that does not exist.

2. The sections not yet reviewed

Four of the five observations express one criterion, that a value stated is a value to be grounded. The remaining sections were scanned against it by rule: a sentence asserting a setting is flagged where neither it nor its neighbour carries a causal clause, a constraint answered, or a citation.

Eleven sentences flag in Section 3. Two are grounded on re-reading, the evaporation sign convention and the sediment thickness taper, which the pattern missed. Nine assert a setting with no ground:

Section	Setting	Value
3.1	Uniform Manning coefficient	0.023 s $m^{-1/3}$
3.1	External forcing update interval	300 s
3.1	Courant limit, step cap, growth factor	0.7, 30 s, 1.1
3.2	Spectral discretisation	24 directions, 24 frequencies, 0.05–1.0 Hz
3.2	Source terms	third generation, Komen, quadruplets, refraction
3.2	Processes disabled	triads, wave setup
3.2	Breaking and bottom friction	$\alpha = 1.0$, $\gamma = 0.73$; JONSWAP 0.067
3.2	Minimum depth	0.05 m
3.2	Coupling exchange interval	3600 s

Six of the nine fall in Section 3.2, where the review stopped. The concentration is not an artefact of the scan: Section 3.2 is a list of SWAN settings, and the only ground it offers anywhere is for the NetCDF format of the communication file. Applied to Sections 4 and 5 the same scan returns reported measurements rather than settings, so the criterion is a Methods one.

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2.1 Points in the same sections that a scan does not reach

The two nested SWAN grids, at 400 m and 100 m, have no stated relation to the unstructured mesh they exchange with. Whether the inner grid resolves the inlets, which are 400 m and 2900 m wide, is the question a reader of Section 3.1 carries into 3.2.

The 3600 s exchange interval sets the time resolution of the whole coupling, against drifter deployments lasting 0.4 to 7.2 hours.

Stationary mode within each exchange is explained mechanically. The ground for it being adequate, that the wave field equilibrates faster than the hourly exchange in a fetch-limited basin, is not given.

Section 3.4 does not say why the vegetation module was used rather than trachytopes, which is the route the earlier literature on this lagoon takes. The reason held in the project record is that the trachytope canopy term does not enter the momentum equation in this build. How much of that belongs in the manuscript is a judgement, but silence invites the question.

Section 3.6 states the exclusion of the suspended sediment contribution to density as a property of the design. A reviewer may read it as a parameter tuned until the runs completed. The evidence against that reading, that the two affected members abort at the same cell and the same instant to six decimal places on independent attempts, is not in the manuscript.

3. The journal's requirements

Two sources carry assessable criteria: Elsevier's reviewer guidance for original research articles, and the *Estuarine, Coastal and Shelf Science* Guide for Authors. The editorial by the journal's Editors-in-Chief, Borja and Elliott (2024), was read and concerns the shortage of referees rather than the assessment of manuscripts. It contributes one fact that bears on this submission: its Table 1 lists a badly written or not understandable paper among the acceptable reasons for a referee to decline an invitation, so clarity affects whether a manuscript is reviewed at all.

3.1 Measurable compliance

Limit	Journal	Manuscript	
Abstract	250 words	238	within
Body text	8000 words	6580	within
Figures	8	8	at the limit
Tables	3	4	**over**
Keywords	1 to 7	6	within
Highlights	3 to 5 bullets, 85 characters	5, all within	within

One table must go, and the smallest is not the cheapest to lose. Table 1 holds the published source for every canopy parameter, which is what supports the claim that nothing was fitted. Table 2, the ensemble members, is reproducible as a sentence plus the member code already defined, and is the least destructive reduction. Figures stand exactly at the limit, so any figure added in revision displaces another.

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3.2 Criterion by criterion

Criterion	State	Exposure
Title describes the content	Names the three factors and the ranking	none
Abstract stands alone, with limitations	Reports the ranking with intervals; carries no limitation	moderate
Introduction current, objectives at the end	22 of 29 references predate 2015; closes on a roadmap paragraph	high (item 3)
Methods replicable	Nine settings ungrounded; no convergence test	high (item 5)
Results well presented	Four tables against three; a deployment-level sensitivity exists and is not reported	moderate
Conclusions supported, limitations emphasised	A Limitations subsection names the uncalibrated erosion threshold	low
References up to date	22 of 29 predate 2015	high (item 3)
Language quality	No professional editing	moderate

3.3 Scope

The Guide for Authors states that the journal does not accept manuscripts on data of mainly local interest, and discourages reports of mainly local and descriptive nature. This is the one criterion on which a rejection would be structural rather than correctable in revision, and it is applied by the editors before referees are invited, on the title, abstract and Introduction alone.

The manuscript's defence is that its claims concern what a class of validation data can certify, and how a factorial behaves when a dominant factor is omitted, neither of which is specific to this basin. That defence currently sits in Sections 5 and 6. Removing the geometry from the Introduction, as observation 2 proposes, acts in the same direction.

3.4 Points arising from the criteria

1. Four tables against a limit of three.
2. The abstract carries no limitation.
3. The Introduction closes on a roadmap rather than on the objectives.
4. The generality claim is absent from the abstract and the Introduction.
5. Language editing before submission.
6. A deployment-level sensitivity is computed and not reported.

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